3-D PLOT

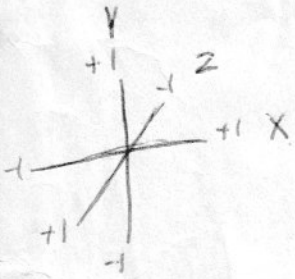
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Specifications 3-D PLOT runs with less than 4K of memory on the TRS-8Ø* Mod I Lev II

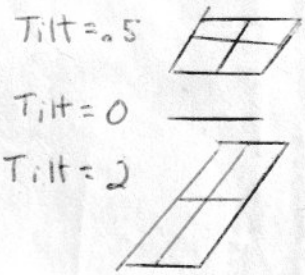
Loading 3-D PLOT loads as a BASIC program with CLOAD"P"

Instructions

Adding the function Functions are added as an equation on line # 2ØØ, with Y on the left and a mathmatical expression usually containing X and Z on the right. (i.e. 2ØØ LET Y=Ø or 2ØØ Y=Z^2 - X^2



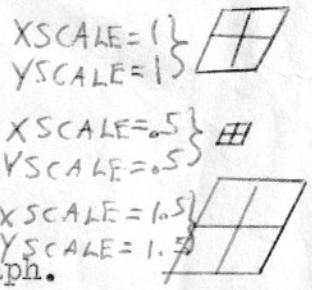
Running Type RUN. Press (ENTER)



The Questions When in doubt just press (ENTER)

TILT? This is the tilt of the grid. standard tilt: .5

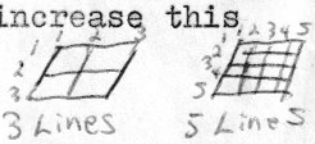
X SCALE?
Y SCALE? This is used to enlarge or reduce the graph's scale. standard SCALE: 1



Switch? Answer NO. standard switch: NO

HOW MANY LINES? This is the number of lines on the graph. If the plot is too cluttered reduce this number, if its too undetailed increase this number. standard lines: 7

Note: # Lines Is Always around number



3-D PLOT will now graph the function. When it is done it will print a D in the upper lefthand corner. Press (BREAK) and type RUN to change perimeters.

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Examples and Explainations

2ØØ Y = Ø

Perimeters: any (experiment!)
This function is recommended to explore the effects of different TILT, SCALES, & LINES

2ØØ Y = Z↑2 - X↑2

Perimeters: standard (just answer by pressing (ENTER))

2ØØ Y = -SQR((Z/2)↑2 + (X/2)↑2)

Perimeters: Tilt: .75 XSCALE:1 YSCALE:1 SWITCH:NO LINES:7
3-D PLOT usually graphs the x axis and z axis from -1 to 1, but by using the terms (X/2) and (Z/2) the graph area becomes -2 to 2. In the same way (X+1) graphs X from Ø to 2 and ((X+1)/2) plots from Ø to 1

2ØØ Y = .3* SWITCH*SQR((Z*2)↑2 +(X*2)↑2 +1)

Perimeters: Tilt .3 XSCALE 1.2 YSCALE 1.2 SWITCH YES LINES 7
The Computer only gives the positive roots of an expression. If both sets of roots are wanted then You should answer the switch question: YES and use SWITCH*SQR instead of just SQR

2ØØ IF X<>Ø OR Z<>Ø THEN Y= -1/ SQR(X↑2*18 + Z↑2*18) + 1 ELSE NO=1

Perimeters: Tilt .3 XSCALE 1.3 YSCALE 1.3 SWITCH NO LINES 9
The IF-THEN-ELSE format can be used to plot functions that sometimes attempt the impossible (like division by zero or sqrare rooting a negative number). Use one of two formats:

IF conditions when alls o.K. THEN function ELSE NO=1
IF conditions that cause error THEN NO=1 ELSE function

2ØØ IF X=Ø AND Z=Ø THEN Y=-1 ELSE Y= -1/SQR(X↑2*18+Z↑2*18)+1

Perimeters: Tilt .3 XSCALE 1.3 YSCALE 1.3 SWITCH NO LINES 9
******** BLACK HOLE ********